

Tour de Lune

2 ① $\boxed{\vec{F}_G = \frac{-Gm_L m_T}{\|TL\|^2} \vec{ur}}$

$$\vec{ur} = \frac{\vec{TL}}{\|\vec{TL}\|}$$

2 ② $\vec{F}_G$ est 1 force centrale

$$\frac{d\vec{L}_T}{dt} = \sum_i \vec{m}_i (\vec{F}_i) = \vec{m}_T (\vec{F}_G) = \vec{0}$$

$$\Rightarrow \vec{L}_T = \text{cte}$$

$$\Rightarrow \vec{OM} \wedge m\vec{v} = \underline{\vec{L}_T(0) = \text{cte}} \quad \rightarrow \text{le mv est plan.}$$

2 ③ $\vec{OM} = r\vec{ur}$

$$\vec{v} = \dot{r}\vec{ur} + r\dot{\theta}\vec{u}_\theta$$

$$\vec{L}_T = \vec{OM} \wedge m\vec{v}$$

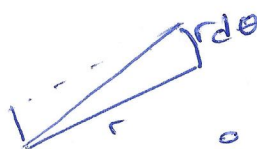
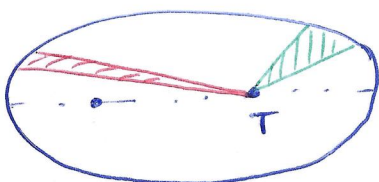
$$= r\vec{ur} \wedge m(\dot{r}\vec{ur} + r\dot{\theta}\vec{u}_\theta)$$

$$\boxed{\vec{L}_T = mr^2\dot{\theta}\vec{u}_z}$$

2 ④ $\|\vec{L}_T\| = \text{cte} \Rightarrow mr^2\dot{\theta} = \text{cte}$

$$\Rightarrow \underline{r^2\dot{\theta} = \text{cte}} \quad \text{ie } C = \text{cte}$$

2 ⑤ Les aires balayées par le rayon vecteur pendant des durées égales sont égales.



$$A = \frac{rd\theta \times r}{2}$$

$$= \frac{r^2}{2} \dot{\theta} dt$$

$$\dot{\theta} = \frac{d\theta}{dt}$$

$$d\theta = \dot{\theta} dt$$

$$= \frac{C}{2} dt$$

$\frac{C}{2}$: vitesse areolaire (cte)

3 ⑥ PFD ref. géocentrique:

$$\Sigma \vec{F} = m \vec{a}$$

$$\vec{F}_G = m \begin{pmatrix} a_r \\ a_\theta \end{pmatrix} \vec{u}_r, \vec{u}_\theta$$

$$\vec{a}_r = (\ddot{r} - r\dot{\theta}^2) \vec{u}_r$$

$$\begin{cases} -\frac{Gm_T m_L}{r^2} = m_L a_r \\ 0 = m a_\theta \end{cases}$$

$\Rightarrow$ l'accél
orthoradiale est
nulle.

2 ⑦

$$\vec{e} = \frac{(\ddot{r} \vec{u}_r + r\ddot{\theta} \vec{u}_\theta) + m r^2 \ddot{\theta} \vec{u}_\theta}{G m_T m_L} - \vec{u}_r$$

$$= \frac{1}{G m_T m_L} \left[-m C \ddot{r} \vec{u}_\theta + m C r \ddot{\theta} \vec{u}_r \right] - \vec{u}_r$$

$$\vec{e} = \frac{1}{G m_T} \left[-C \ddot{r} \vec{u}_\theta + \frac{C^2}{r} \vec{u}_r \right] - \vec{u}_r$$

$$C = r^2 \dot{\theta}$$

$$\frac{C}{r} = r \dot{\theta}$$

2 ⑧

$$\frac{d\vec{e}}{dt} = \frac{1}{G m_T} \left[-C \ddot{r} \vec{u}_\theta + C r \ddot{\theta} \vec{u}_r + \frac{C^2}{r} \ddot{\theta} \vec{u}_\theta \right] - \ddot{\theta} \vec{u}_\theta$$

$$= \frac{1}{G m_T} \left[\underbrace{r^2 \ddot{\theta}}_{\frac{G m_T}{r^2}} \left[-\ddot{r} + r \ddot{\theta}^2 \right] \vec{u}_\theta - \ddot{\theta} \vec{u}_\theta \right]$$

$$-\frac{C^2}{r^2} \ddot{r} \vec{u}_r$$

$$\frac{d\vec{u}_\theta}{dt} = -\dot{\theta} \vec{u}_r$$

$$\frac{d\vec{e}}{dt} = \dot{\theta}\vec{u}_\theta - \dot{\theta}\vec{u}_\theta = \underline{\vec{0}}$$

Ce qui prouve que ce vecteur est constant.

2 (9)
$$E_p = - \frac{G M_L M_T}{r}$$

2 (10)
$$E_c = \frac{1}{2} m v^2 = \frac{1}{2} m (\dot{r}^2 + (r\dot{\theta})^2)$$

$$= \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2$$

$C = r^2 \dot{\theta}$
 $C^2 = r^4 \dot{\theta}^2$

$$E_c = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m \frac{C^2}{r^2}$$

3 (11)
$$E_m = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m \frac{C^2}{r^2} - \frac{G M_L M_T}{r}$$

TEM appliqué à la Lune

$$\Delta E_m = \sum_i W(\vec{F}_i, \vec{nc}) = 0. \quad \text{car } \vec{F}_G \text{ est conservative}$$

2 (12) A l'apogée et au périée $\dot{r} = 0$

$$E_m = \frac{1}{2} m \frac{C^2}{r^2} - \frac{G M_L M_T}{r}$$

r_a et r_s sont
les 2 racines
de cette eq^e du
2^d degré

$$r_+ + r_- = -\frac{b}{a}$$

$\Rightarrow$